Data storage and management

Vladislav Goncharenko

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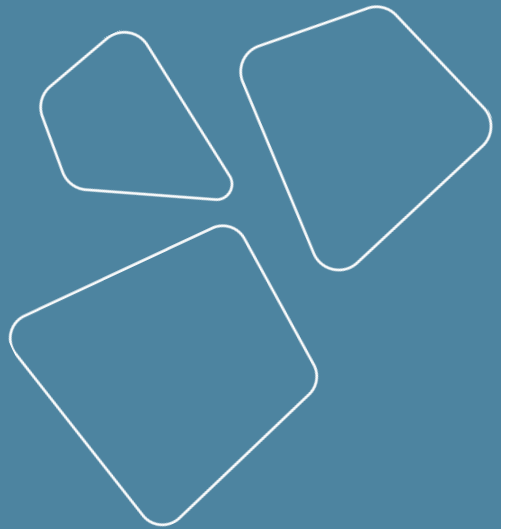
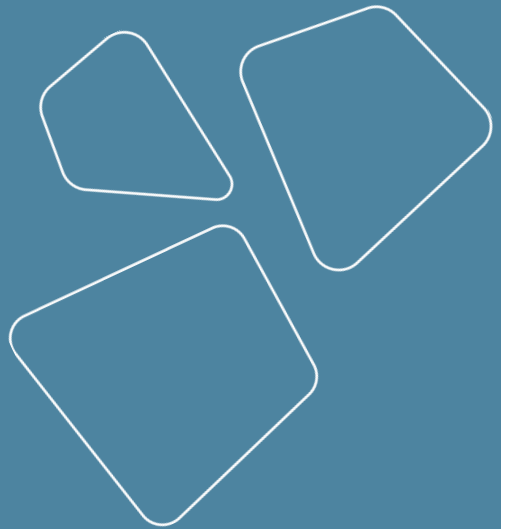
Recap



- Code storages
- Git servers
- Team workflow
  - code distribution
- CI/CD
- Code quality tools
  - formatters
  - imports manager
  - linters
- Documentation generation
- pre-commit

Outline



- Data sources
- Data storage
  - small data
  - big data
- Data engineering
  - ETL and ELT
  - datahub
- Experiments parametrization
  - Hydra
  - Vault

Data sources

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01

Data sources

- Primary
  - keyboard input
  - web projects
    - typical architecture: kafka - hadoop - airflow - ml
    - see courses on web and big data
  - photo and video cameras
    - self driving cars
  - many other sensors
    - temperature, humidity, illumination
- Secondary
  - results of sql queries
  - speech to text
  - web scraping
  - LLM
  - synthetic data
    - 3d modelling
  - etc...

Characteristics of sources

Factor	Primary Data	Secondary Data
Definition	Primary Data refers to the first-hand data collected by the team. It is collected based on the researcher's needs.	Secondary Data has been collected by other teams in the past. It does not necessarily need to be aligned with the researcher's requirements.
Data	Real-time Data	Historical Data
Process	Time Consuming	Quick and Easy
Cost	Expensive	Economical
Collection Time	Long	Short
Available In	Raw and Crude form	Refined form
Accuracy and Reliability	Very high	Relatively less
Examples	Personal Interviews, Surveys, Observations, etc.	Websites, Articles, Research Papers, Historical Data, etc.

Kinds of data

- Structured
  - aka externally structured
  - tabular or relational
- Unstructured (internally structured)
  - Texts
  - Images
  - Video
  - Voice
  - Graphs
  - ....

This naming is a bit weird but this is how it works now.

Kinds of data

Structured Data	Unstructured Data
Structured Data is organized and has a fixed and defined schema.	Unstructured Data does not have any pre-defined structure to it.
Structured data is **quantitative** data that consists of **numbers and values**.	Unstructured data is **qualitative** data that consists of **text, images, audio, video,** etc.
Structured Data is stored in **Relational Databases** and **Data warehouses**.	Unstructured Data generally is stored in **Data Lakes.**
Structured Data is stored in tabular formats such as **SQL databases.**	It is typically stored in **NoSQL databases**.
It is collected from sensors, network logs, web server logs, OLTP systems, etc.	It is sourced from email messages, word-processing documents, pdf files, etc.
It requires **less storage** and is **highly scalable.**	It needs **more storage** space and is **difficult to scale**.

Data storages

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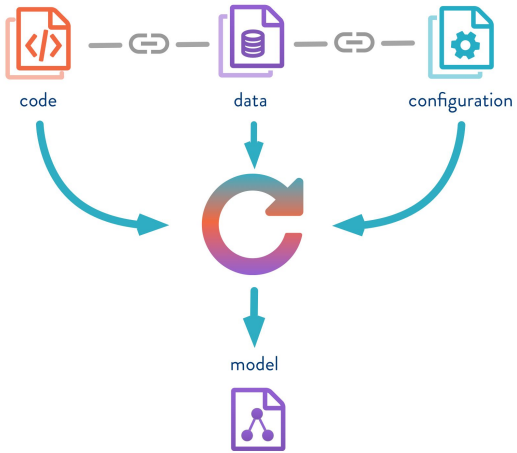
Ways to store data

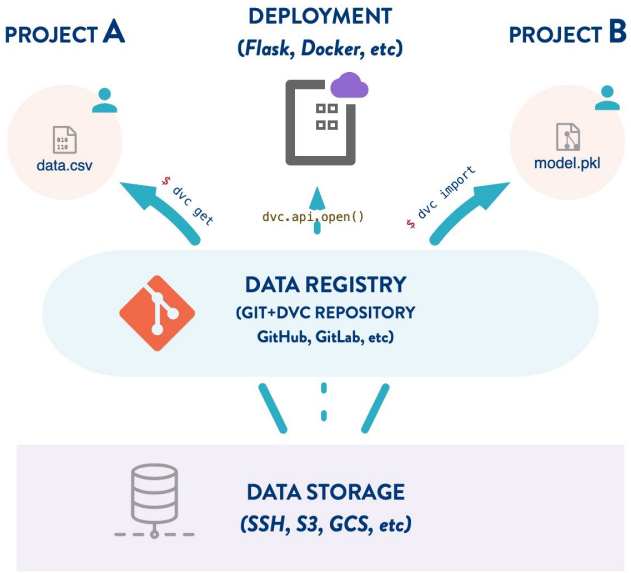
Relevant both for datasets and trained model artifacts

- Locally
- Remotely
- Distributed
  - small data
    - DVC
    - git LFS
  - big data
    - MapReduce paradigm
    - e.g. hadoop stack or YTsaurus

Data Version Control (DVC)

- git for data is DVC
- tutorials: one, two





Other data versioning systems

- https://lakefs.io/
- https://www.atlassian.com/git/tutorials/git-lfs

https://alternativeto.net/software/dvc/

lakeFS

Small data vs big data

1. Cost of infrastructure

   for small data VPS (ssh), s3 storage or GDrive

   for big data - need MapReduce (3 VPS for hadoop at least)
2. Tracking of data state

   for small data DVC or git LFS stores full snapshot of data

   for big data - you don’t care, also it’s not possible

Data storages

- https://github.com/feast-dev/feast
- https://github.com/NVIDIA/aistore
- https://github.com/qdrant/qdrant

FEAST

drant

Data Engineering

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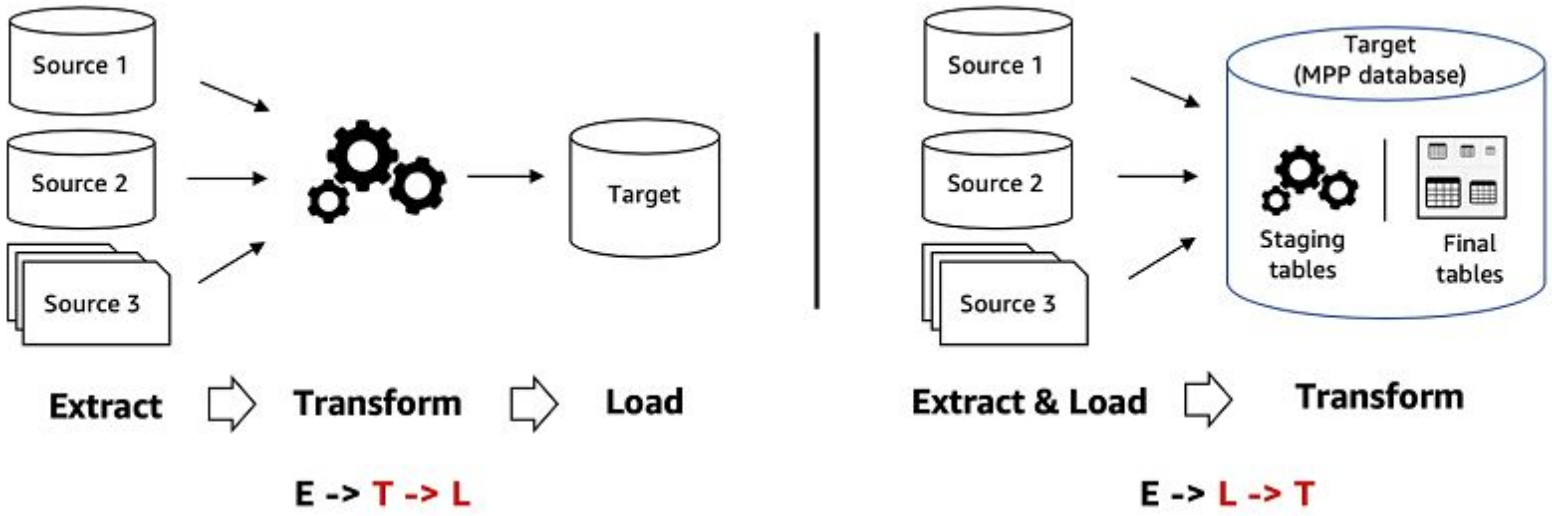
03

ETL and ELT

ETL: extract, transform, load

ELT: extract, load, transform



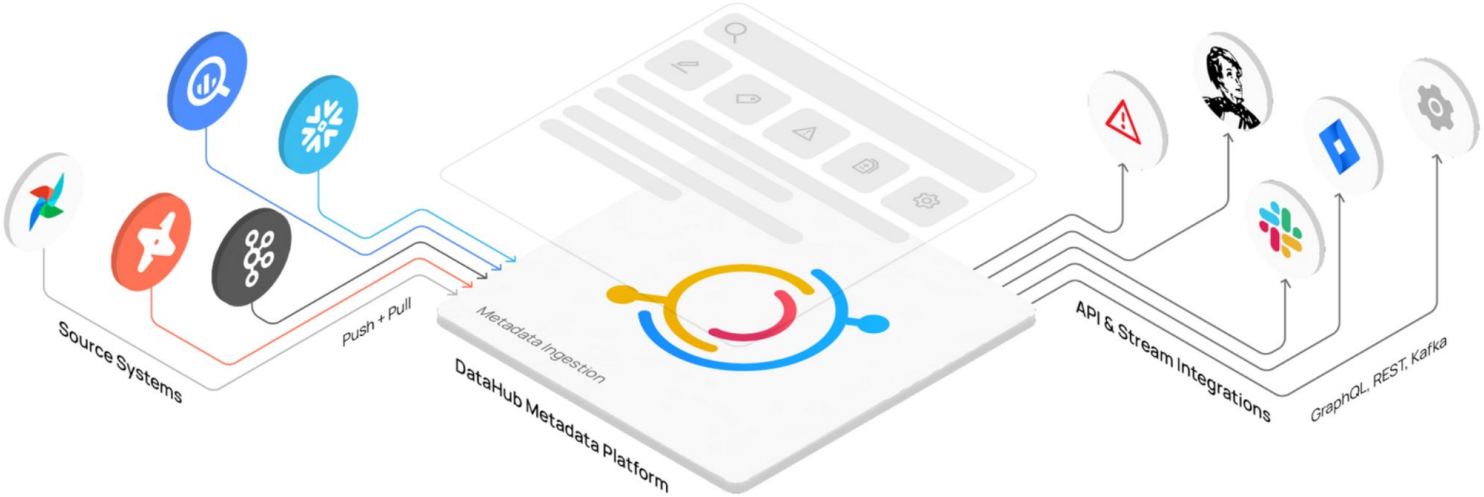
ETL vs ELT

ELT	ETL
ELT tools do not require additional hardware	ETL tools require specific hardware with their own engines to perform transformations
Mostly Hadoop or NoSQL database to store data.Rarely RDBMS is used	RDBMS is used exclusively to store data
As all components are in one system, loading is done only once	As ETL uses staging area, extra time is required to load the data
Time to transform data is independent of the size of data	The system has to wait for large sizes of data. As the size of data increases, transformation time also increases
It is cost effective and available to all business using SaaS solution	Not cost effective for small and medium business
The data transformed is used by data scientists and advanced analysts	The data transformed is used by users reading report and SQL coders
Creates ad hoc views.Low cost for building and maintaining	Views are created based on multiple scripts.Deleting view means deleting data
Best for unstructured and non-relational data. Ideal for data lakes. Suited for very large amounts of data	Best for relational and structured data. Better for small to medium amounts of data

Data Hub

Single place to store meta information for all data

https://datahubproject.io/



Database modelling

e.g. Data vault modeling

Experiments parametrization

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Written and run model - what’s next?

- Produce the best model in one run - unrealistic
- It requires a large number of experiments
- How not to get confused?
- What if there are several people?
- And then there's rolling into production...

Possible solutions

- Copy and paste - already passed level
- Making git branches for each experiment is impractical
- Use configuration files!

Configuration management systems

- https://docs.pydantic.dev/latest/
- https://github.com/HBNetwork/python-decouple
- https://github.com/dynaconf/dynaconf
- https://github.com/facebookresearch/hydra - made for ML research
- etc...

Config management for ML

https://github.com/facebookresearch/hydra

Example of app

- Config is defined by yaml or dataclass
- Functions are wrapped to hydra.main
- Choose config in CLI
- Saves final version of config
- Example of app refactoring https://github.com/ArjanCodes/2021-config/

my_app_type_error.py

```python
from dataclasses import dataclass

import hydra
from hydra.core.config_store import ConfigStore

@dataclass
class MySQLConfig:
    host: str = "localhost"
    port: int = 3306

cs = ConfigStore.instance()
# Registering the Config class with the name 'config'.
cs.store(name="config", node=MySQLConfig)

@hydra.main(version_base=None, config_name="config")
def my_app(cfg: MySQLConfig) -> None:
    # pork should be port!
    if cfg.pork == 80:
        print("Is this a webserver?!")

if __name__ == "__main__":
    my_app()
```

YAML to store values

Supports:

- default values
- variables
- hierarchical configs

```yaml
defaults:
  - files: mnist
  - _self_
paths:
  log: ./runs
  data: ${hydra:runtime.cwd}/../data/raw
params:
  epoch_count: 20
  lr: 5e-5
  batch_size: 128
```

Hierarchical configs

```python
@dataclass
class MySQLConfig:
    host: str = "localhost"
    port: int = 3306

@dataclass
class UserInterface:
    title: str = "My app"
    width: int = 1024
    height: int = 768

@dataclass
class MyConfig:
    db: MySQLConfig = field(default_factory=MySQLConfig)
    ui: UserInterface = field(default_factory=UserInterface)

cs = ConfigStore.instance()
cs.store(name="config", node=MyConfig)

@hydra.main(version_base=None, config_name="config")
def my_app(cfg: MyConfig) -> None:
    print(f"Title={cfg.ui.title}, size={cfg.ui.width}x{cfg.ui.height} pixels")
```

Hydra without main decorator

The Compose API is useful when @hydra.main() is not applicable. For example:

- Inside a Jupyter notebook
- In parts of your application that does not have access to the command line
- If you want to compose multiple configuration objects

```python
from hydra.experimental import compose, initialize
from omegaconf import OmegaConf

if __name__ == "__main__":
    # context initialization
    with initialize(config_path="conf", job_name="test_app"):
        cfg = compose(config_name="config", overrides=["db=mysql", "db.user=me"])
        print(OmegaConf.to_yaml(cfg))

    # global initialization
    initialize(config_path="conf", job_name="test_app")
    cfg = compose(config_name="config", overrides=["db=mysql", "db.user=me"])
    print(OmegaConf.to_yaml(cfg))
```

Advanced syntax

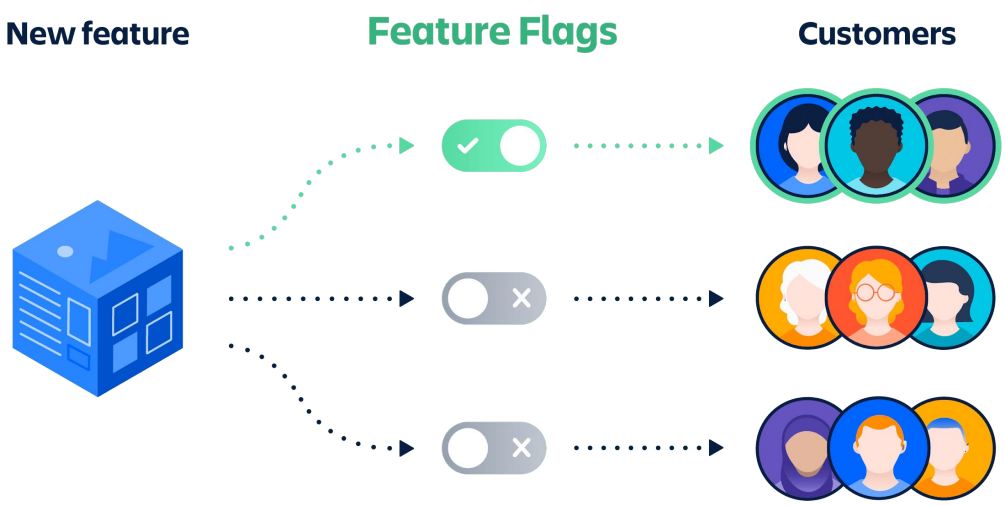
```
defaults:
  - models/catboost: master
  - models/optuna: catboost
  - db_client@client
  - pools: samples
  - pools/processors: common_proc

target: kp
validation: online
version: 3
verbose: true

predictions: predicted_${.target}_on_${.validation}_v${.version}.csv

```

Integrated with popular libraries

- hydra-zen: vanilla wrapper over Hydra
- lightning-hydra-template: template for popular training library
- hydra-torch: wrapper for most popular neural deep learning library

Using configs for AB testing

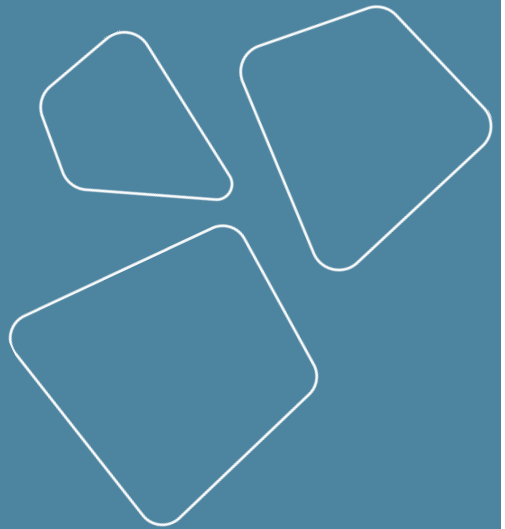
Using configs (e.g. when starting Docker container/pod) we can build infrastructure for AB testing or other needs.



Special case - secrets

Default solution - Vault https://github.com/hashicorp/vault

Revise



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Thanks for attention!

Questions?